Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-II-Q02] | Part A (Q1-Q5)

Answer: 294

1. Each of the three pairwise squared distances is 98, so the vertices form an equilateral triangle.

2. In a cube, three mutually equidistant vertices cannot all be joined by edges (the cube graph has no triangles) or by space diagonals (each vertex has only one opposite). They must be joined by face diagonals. Hence $2s^2 = 98$ and surface area $6s^2 = 294$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q05] | Part A (Q1-Q5)

Answer: 505

1. Decompose by the nearest box feature. The box contributes 60, and the six face slabs contribute $2(12+15+20)=94$.

2. The twelve edge quarter-cylinders combine into cylinders of total axial length $3+4+5=12$, contributing 12π. The eight corner octants form one sphere of volume $4\pi/3$.

3. Total volume is $154+40\pi/3=(462+40\pi)/3$, giving 505.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q04] | Part A (Q1-Q5)

Answer: 028

1. With centroid $G = (A + B + C + D)/4$, the center opposite A is $(B + C + D)/3 = G - (A - G)/3$. The same relation holds at every vertex.

2. Thus the face-center tetrahedron is a homothetic image with scale $-1/3$, giving volume ratio $1/27$. The requested sum is 28.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q05] | Part A (Q1-Q5)

Answer: 216

1. Over the radius-six disk, the wedge height is an affine function, zero at the tangent contact point and twelve at its diameter-opposite point.

2. Opposite disk points have heights summing to twelve, so the average height is six. Volume $= \pi \cdot 6^2 \cdot 6 = 216\pi$, giving 216.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q05] | Part A (Q1-Q5)

Answer: 014

1. The base contact travels a circle of radius equal to the slant height $\ell = \sqrt{r^2 + h^2}$. One circuit rolls distance $2\pi\ell$, equal to seventeen base circumferences $17 \cdot 2\pi r$.

2. Thus $\ell = 17r$ and $h^2 = 288r^2$, yielding $h/r = 12\sqrt{2}$ and answer 14.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2012-II-Q05] | Part A (Q1-Q5)

Answer: 750

1. Each flat triangular face has altitude $20 - 15/2 = 25/2$, which becomes the pyramid slant height. The base apothem is $15/2$.

2. The vertical height is $\sqrt{(25/2)^2 - (15/2)^2} = 10$. Thus the volume is $15^2 \cdot 10/3 = 750$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2016-I-Q04] | Part A (Q1-Q5)

Answer: 108

1. Let U, V be the two base neighbors of A, and W its vertical neighbor. The two relevant planes meet along UV. In isosceles triangle AUV, $AU = AV = 12$ and angle $UAV = 120^\circ$.

2. The midpoint M of UV satisfies $AM = 12\cos 60^\circ = 6$. Both AM and WM are perpendicular to UV, so angle AMW is the given dihedral angle.

3. Right triangle AMW gives $h = 6\tan 60^\circ = 6\sqrt{3}$, and $h^2 = 108$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2017-I-Q04] | Part A (Q1-Q5)

Answer: 803

1. Equal apex distances put its perpendicular projection at the base circumcenter. The base altitude is $\sqrt{20^2 - 12^2} = 16$, so area $K = 192$ and circumradius $R = 20 \cdot 20 \cdot 24/(4K) = 25/2$.

2. The pyramid height is $\sqrt{25^2 - R^2} = 25\sqrt{3}/2$. Its volume is $Kh/3 = 800\sqrt{3}$, giving $m + n = 803$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2000-I-Q08] | Part B (Q6-Q10)

Answer: 052

1. The initial liquid occupies $(9/12)^3 = 27/64$ of the total volume. On reversal, the empty apex cone occupies $37/64$.

2. Its height is $12\sqrt[3]{37/64} = 3\sqrt[3]{37}$. Subtracting from 12 gives the liquid depth, so the requested sum is $12 + 3 + 37 = 52$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q06] | Part B (Q6-Q10)

Answer: 348

1. Triangles have only three squared-side patterns: $(1,1,2)$, $(2,2,2)$, and $(1,2,3)$. The first occurs four times per face, or 24 times, with area $1/2$.

2. The equilateral pattern occurs 8 times with area $\sqrt{3}/2$. The last occurs once for each space diagonal and each of the other six vertices, or 24 times, with area $\sqrt{2}/2$. The counts sum to $56 = \binom{8}{3}$.

3. Total area is $12 + 4\sqrt{3} + 12\sqrt{2} = 12 + \sqrt{48} + \sqrt{288}$, giving 348.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2005-II-Q10] | Part B (Q6-Q10)

Answer: 011

1. Take octahedron vertices $(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)$. Two square pyramids of base area two and height one give volume $4/3$.

2. Face centers are $(\pm 1/3, \pm 1/3, \pm 1/3)$, so cube volume is $(2/3)^3 = 8/27$. Their ratio is $9/2$, giving 11.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2012-I-Q08] | Part B (Q6-Q10)

Answer: 089

1. The cutting plane through $(0,1,0),(1/2,0,0),(1,1,1/2)$ has equation $2x+y-4z=1$. Its smaller portion is near the bottom edge BC.

2. Extend DM to $K=(1,-1,0)$, so $CK=2$ and B is the midpoint of CK. The tetrahedron $KDCN$ has base DCN of area $1/4$ and perpendicular height two, so volume $1/6$.

3. The part beyond the cube is a similar tetrahedron at K of linear scale $1/2$, volume $1/48$. Hence the smaller cube portion is $1/6-1/48=7/48$.

4. The larger portion is $41/48$, giving $p+q=89$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-II-Q09] | Part B (Q6-Q10)

Answer: 384

1. Near its lowest vertex, a plane perpendicular to the cube body diagonal cuts an equilateral triangle. At the rim, its circumradius is 4, so its side length is $4\sqrt{3}$.

2. If the equal intercepts along the three mutually perpendicular cube edges are t, the section side is $t\sqrt{2}$, giving $t = 2\sqrt{6} < 8$. Thus the submerged part is exactly a corner tetrahedron, of volume $t^3/6 = 8\sqrt{6}$.

3. Its depth is $t/\sqrt{3} = 2\sqrt{2} < 10$, so it does not reach the barrel bottom. The displaced volume is the tetrahedron volume, and $v^2 = 384$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2021-I-Q06] | Part B (Q6-Q10)

Answer: 192

1. Set $A = 0$, with B, C, D on perpendicular coordinate axes at distance s, and $G = (s, s, s)$. Expanding distances to $P = (x, y, z)$ gives $PB^2 + PC^2 + PD^2 - PG^2 = 2PA^2$; both linear coordinate terms and s^2 cancel.

2. Therefore $2PA^2 = 36000 + 18000 + 28800 - 9072 = 73728$. Thus $PA = \sqrt{36864} = 192$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q12] | Part C (Q11-Q15)

Answer: 101

1. Each face becomes three corner triangles and three outer tetrahedron faces, all of half side length. Thus at stage i there are $4 \cdot 6^i$ faces, with side scale 2^{-i}.

2. The attached tetrahedra in the next step have scale $2^{-(i+1)}$, hence volume $8^{-(i+1)}$ each. The increment is $\frac{1}{2}(3/4)^i$.

3. The first three additions give total volume $1 + 1/2 + 3/8 + 9/32 = 69/32$. The requested sum is 101.

4. The required non-overlap through three stages can be checked exactly: start with vertices $(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)$. Each outward apex and face midpoint remains rational. Separating-axis tests on the resulting 173 tetrahedra and coplanar-triangle tests on the 864 final boundary faces show only boundary contact, with no positive-volume or hidden positive-area overlap. The accompanying executable certificate uses exact integer projections, so the stated face count and additive volume are valid through stage three.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q15] | Part C (Q11-Q15)

Answer: 417

1. Take $A=(0,0,0), E=(8,8,8)$ and $I=(6,8,8), J=(8,6,8), K=(8,8,6)$. The parallel tunnel lines from I, J, K reach $L=(0,2,2), M=(2,0,2), N=(2,2,0)$.

2. The wall through IJ has plane $2z=x+y+2$ and meets the z-axis at $O=(0,0,1)$. Its pentagon is rectangle $ILMJ$ plus triangle LOM. Their areas are $(2\sqrt{2})(6\sqrt{3})=12\sqrt{6}$ and $(2\sqrt{2})(\sqrt{3})/2=\sqrt{6}$, totaling $13\sqrt{6}$.

3. Each of the three faces at E loses a legs-2 right triangle, area 2. Each face at A loses a quadrilateral, for example $(0,0), (1,0), (2,2), (0,1)$, also area 2.

4. The total is $6(64-2)+3(13\sqrt{6})=372+39\sqrt{6}$. Thus $m+n+p=417$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-I-Q15] | Part C (Q11-Q15)

Answer: 163

1. Use $D=(0,0,0), C=(12,0,0), B=(12,12,0), A=(0,12,0)$. Equal distances to C, D put E in $x=6$. Its distance to the filled square forces its perpendicular foot to be within the square; otherwise its squared distance to D and the nearest edge contradict $DE=14$ and distance 12. Hence, choosing the upper side, $E=(6,4,12)$.

2. Isosceles trapezoid symmetry gives $G=(3,y,z), F=(9,y,z)$. The face $ADEG$ lies in $z=2x$, so $z=6$. Then $AG=8$ yields $(y-12)^2=19$.

3. Therefore $EG^2=3^2+6^2+(8\pm\sqrt{19})^2=128\pm16\sqrt{19}$. The specified minus form selects $128-16\sqrt{19}$, so the required sum is 163.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q11] | Part C (Q11-Q15)

Answer: 512

1. The full cone has slant height 5, lateral area 15π and total painted area 24π. At similarity scale t, the small cone retains painted area $15\pi t^2$, while its new cut face is unpainted.

2. Equate $15t^2/(24-15t^2) = t^3/(1-t^3)$. Cancellation gives $15 = 24t$, so $t = 5/8$.

3. The common ratio is $125/(512-125) = 125/387$, giving 512.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q14] | Part C (Q11-Q15)

Answer: 813

1. Unroll the wall and extend the tangent free portion in that development. Tautness makes the combined path straight, rising 4 over length 20, so its total horizontal projection is $\sqrt{400-16}=8\sqrt{6}$.

2. The endpoint is at horizontal radius 12. Its tangent to the radius-eight circle has projected length $\sqrt{12^2-8^2}=4\sqrt{5}$. The constant length-to-projection ratio makes the free rope length $20(4\sqrt{5})/(8\sqrt{6})=5\sqrt{30}/3$.

3. The contact length is $(60-\sqrt{750})/3$, giving $60+750+3=813$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-II-Q11] | Part C (Q11-Q15)

Answer: 625

1. The slant height is 800, so its lateral surface develops to a radius-800 sector with angle $360^\circ(600/800) = 270^\circ$. Opposite generators are separated by 135° in the development.

2. The straight segment in that sector is the shortest path. Its squared length is $125^2 + (375\sqrt{2})^2 - 2(125)(375\sqrt{2})\cos 135^\circ = 390625$.

3. The distance is 625.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2007-I-Q13] | Part C (Q11-Q15)

Answer: 080

1. Take base $A=(0,0,0), B=(4,0,0), C=(4,4,0), D=(0,4,0)$ and apex $E=(2,2,2\sqrt{2})$. The cutting plane is $x+y+2\sqrt{2}z=6$.

2. Its section has vertices $(1,1,\sqrt{2}), (7/2,1/2,\sqrt{2}/2), (4,2,0), (2,4,0), (1/2,7/2,\sqrt{2}/2)$. Their xy projections have shoelace area eight.

3. Projection onto xy scales area by $1/\sqrt{1+(1/(2\sqrt{2}))^2+(1/(2\sqrt{2}))^2}=2/\sqrt{5}$. Thus the section area is $8\sqrt{5}/2=\sqrt{80}$, giving $p=80$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q15] | Part C (Q11-Q15)

Answer: 871

1. At one corner use $O=(0,0)$ and $P=(a,0)$ with $a=\sqrt{17}$. The inner cut vertex is $R=(t,t)$. The angle condition makes triangle OPR have angles $45^\circ, 105^\circ, 30^\circ$, so $PR=a\sqrt{2}$ and $t=PR\sin 75^\circ=a(\sqrt{3}+1)/2$.

2. The fold through R is horizontal. Its perpendicular foot above P is $S=(a,t)$, so the panel width is $SP=t$. After folding and taping, symmetry puts P vertically over (a,a), while its distance from S remains t. The horizontal separation is $t-a$.

3. Therefore height squared is $t^2-(t-a)^2=a^2\sqrt{3}=17\sqrt{3}$. Height is $\sqrt[4]{867}$; $867=3\cdot 17^2$ has no fourth-power factor. The answer is 871.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2010-I-Q11] | Part C (Q11-Q15)

Answer: 365

1. The region is a triangle with apex $C = (8, 10)$ and base endpoints $A = (9/2, 13/2), B = (39/4, 33/4)$ on the rotation axis.

2. The perpendicular foot of C is $(87/10, 79/10)$, lying within AB. Therefore rotation gives two cones sharing a base, with total axial height $AB = 7\sqrt{10}/4$ and base radius $7/\sqrt{10}$.

3. The volume is $\pi(49/10)(7\sqrt{10}/4)/3 = 343\pi/(12\sqrt{10})$. Hence the answer is $343 + 12 + 10 = 365$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-I-Q15] | Part C (Q11-Q15)

Answer: 053

1. The base chord lies at distance $6\cos 60^\circ = 3$ from the base center. Central symmetry sends it to a parallel chord on the other base at opposite signed offset. The section projects onto the width-6 central strip of the radius-6 disk.

2. Each excluded circular cap has area $12\pi - 9\sqrt{3}$, from its 120° sector minus the central triangle. The projected strip area is $36\pi - 2(12\pi - 9\sqrt{3}) = 12\pi + 18\sqrt{3}$.

3. The plane rises 8 across horizontal distance 6, so the area enlargement factor is $\sqrt{6^2 + 8^2}/6 = 5/3$. Hence section area $20\pi + 30\sqrt{3}$, giving $a + b + c = 53$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2023-I-Q13] | Part C (Q11-Q15)

Answer: 125

1. For three generating edge vectors u, v, w, the rhombus diagonal relations give $|u|^2 = |v|^2 = |w|^2 = (21 + 31)/4 = 13$. Every pairwise dot product is $\pm(31 - 21)/4 = \pm 5/2$.

2. With $x = u \cdot v, y = u \cdot w, z = v \cdot w$, the Gram determinant is $V^2 = 13^3 - 13(x^2 + y^2 + z^2) + 2xyz$. The squared terms are fixed; only the sign of xyz changes.

3. The two squared volumes are $2197 - 975/4 \pm 125/4$, namely $7938/4$ and $7688/4$. Their ratio gives $V_L/V_S = \sqrt{3969/3844} = 63/62$.

4. The fraction is reduced, so $m + n = 125$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2023-II-Q14] | Part C (Q11-Q15)

Answer: 751

1. Use cube-edge coordinates $A=(0,0,0), B=(s,0,0), C=(0,s,s), D=(s,s,s)$ and unit vertical vector u. The vertical rectangle has normal $(0,1,-1)$, so $u_2=u_3$. Heights give $su_1=2, su_2=su_3=4$, whence $s^2=2^2+4^2+4^2=36$.

2. Thus the cube is $[0,6]^3$ and height is $(x+2y+2z)/3$. Reflect all coordinates about 3: the empty region above height 7 has the same volume as $x+2y+2z<9$ inside the cube.

3. The full first-octant tetrahedron has intercepts $9, 9/2, 9/2$ and volume $243/8$. Only the constraint $x\le 6$ truncates it; the removed tetrahedron has intercepts $3, 3/2, 3/2$ and volume $9/8$. Thus air occupies $234/8=117/4$.

4. The water volume is $6^3-117/4=747/4$, giving $m+n=751$.

Source / 来源

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2024-I-Q14] | Part C (Q11-Q15)

Answer: 104

1. Embed the tetrahedron in a rectangular box using alternate vertices. Box dimensions a, b, c satisfy $a^2 + b^2 = 41, a^2 + c^2 = 80, b^2 + c^2 = 89$, giving $(a, b, c) = (4, 5, 8)$. Its volume is $abc/3 = 160/3$.

2. Every face has squared sides $41, 80, 89$. The squared-area identity gives $16F^2 = 2(41 \cdot 80 + 80 \cdot 89 + 89 \cdot 41) - (41^2 + 80^2 + 89^2) = 12096$, so $F = 6\sqrt{21}$.

3. Decomposing into four pyramids gives $V = r(4F)/3$. Hence $r = 20\sqrt{21}/63$, so $m + n + p = 104$.

Source / 来源